

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 46

Delegate IAM Access Analyzer and Detect Organization External Access

IAM Access Analyzer | SecurityAudit | Security Hub | Data Perimeter

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	August 4, 2026
STATUS	Complete - Findings Review Pending

Executive Summary

Logged into the **AWS Organizations management account**, configured the **SecurityAudit** account as the delegated administrator for **IAM Access Analyzer**, and then switched to SecurityAudit to create **external access analyzers** using the **current organization** as the zone of trust. Analyzers were configured in **us-west-2 (Oregon)** and **us-east-1 (N. Virginia)**. The analyzer generated active findings, including a resource analysis view showing **17 resources with active findings**, and the existing Security Hub notification path sent an email alert.

OPERATIONAL POINT

Access Analyzer is a regional detection service. Organization-wide trust scope does not make the analyzer global; it must be created in every region where protected resources are used.

Business Problem

The prior S3 labs intentionally created public ACL and bucket-policy exposure. Preventive controls should not be deployed blindly because broad denies can break legitimate sharing. IAM Access Analyzer provides the detection layer needed to inventory public and cross-account resource exposure before remediation and data-perimeter enforcement.

Objectives

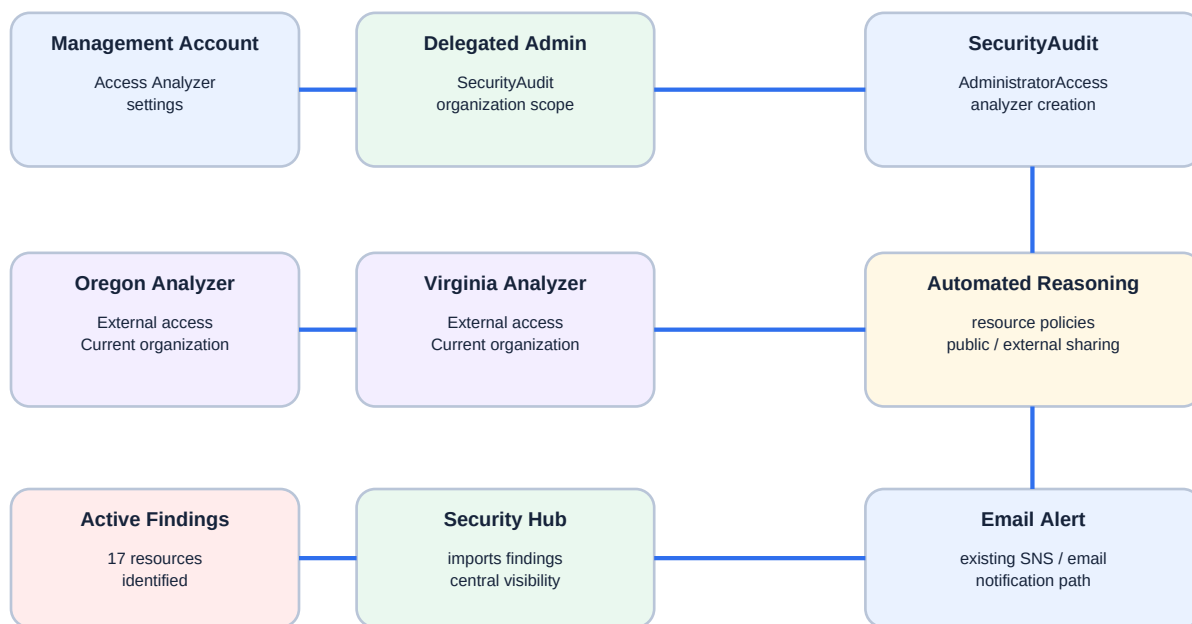
- * Sign into the management account with AdministratorAccess.
- * Open IAM Access Analyzer settings.
- * Delegate Access Analyzer administration to the SecurityAudit account.
- * Sign into SecurityAudit with AdministratorAccess.
- * Create an external access analyzer in us-west-2.
- * Set the zone of trust to the current AWS organization.
- * Create a matching external access analyzer in us-east-1.
- * Review active external-access findings.
- * Validate that findings are forwarded through Security Hub to email.

Environment

Cloud Provider	Amazon Web Services
Management Context	AWS Organizations management account / AdministratorAccess
Delegated Administrator	SecurityAudit account
Primary Service	IAM Access Analyzer

Analyzer Type	Resource analysis - External access
Zone of Trust	Current organization
Region 1	us-west-2 / United States (Oregon)
Region 2	us-east-1 / United States (N. Virginia)
Finding State	Active findings generated
Observed Count	17 resources with active findings in supplied evidence
Notification Path	Access Analyzer -> Security Hub -> existing SNS/email alerting
Status	Complete - exposure review and remediation follow-up pending

Architecture



Access Analyzer detects external sharing outside the organization zone of trust.

Figure 1 - Delegated, multi-region IAM Access Analyzer detection workflow.

Implementation Summary

1. Signed into the AWS Organizations management account.
2. Opened IAM Access Analyzer and navigated to Analyzer settings.
3. Added the SecurityAudit account as delegated administrator.

4. Switched to the SecurityAudit account with AdministratorAccess.
5. Opened IAM Access Analyzer in us-west-2.
6. Created a resource analysis analyzer using the External access finding type.
7. Selected Current organization as the zone of trust.
8. Created the analyzer and allowed it to scan supported resources.
9. Changed to us-east-1 and created a second analyzer with the same external-access and organization-scope settings.
10. Returned to the analyzer findings and confirmed active external-access results.
11. Confirmed the existing Security Hub alerting workflow sent an email notification.

Control and Detection Map

Delegated administration	SecurityAudit account	Moves analyzer operations out of the management account and centralizes security review.
External access analyzer	Resource analysis	Identifies supported resources shared publicly or outside the selected zone of trust.
Zone of trust	Current organization	Treats organization principals as trusted and highlights access from outside the organization.
Regional deployment	Oregon and N. Virginia	Covers the two regions permitted and used by the lab organization.
Security Hub integration	Existing centralized configuration	Aggregates Access Analyzer findings with other security findings.
Email notification	Existing EventBridge/SNS path	Provides analyst notification after findings reach the security pipeline.

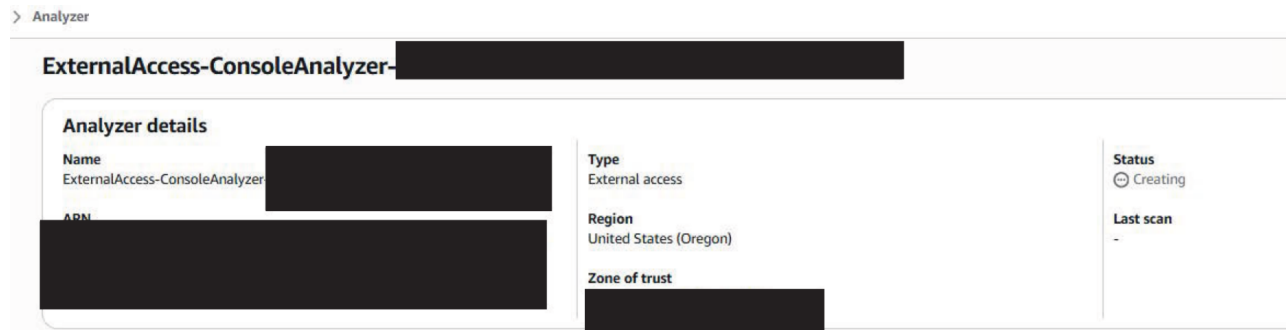
Evidence

Figure 2 - SecurityAudit Configured as Delegated Administrator



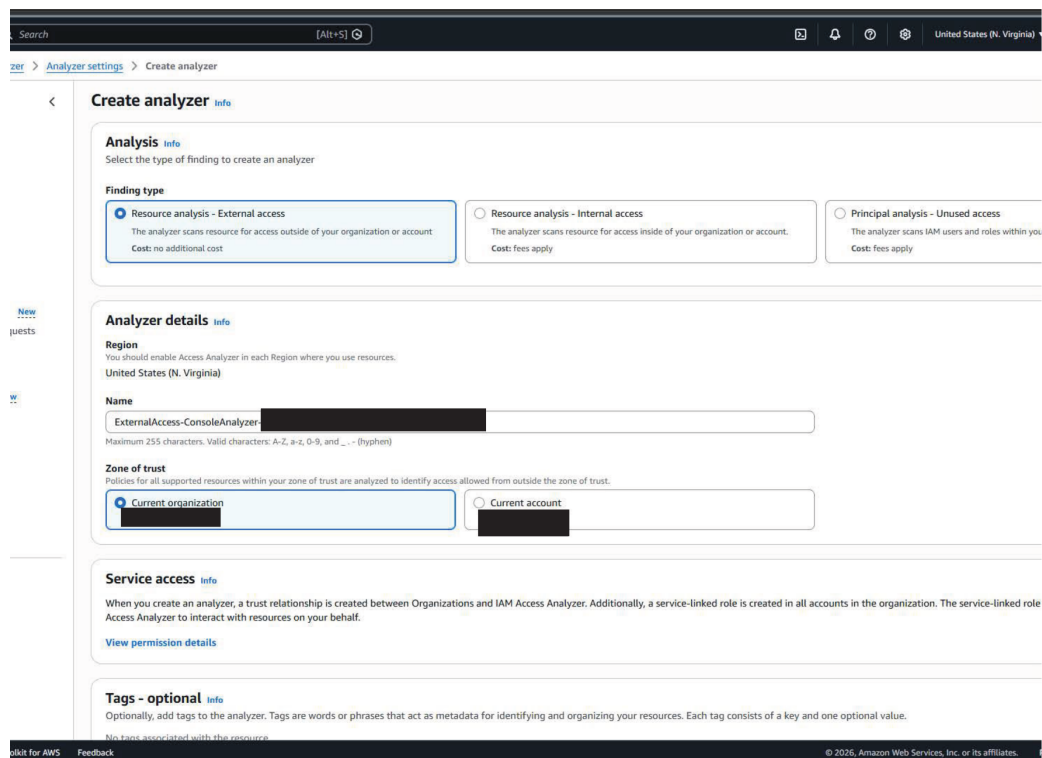
The Access Analyzer settings evidence shows a redacted organization management account and a redacted delegated administrator identified as the current account after switching into SecurityAudit.

Figure 3 - External Access Analyzer Created in Oregon



The analyzer details evidence shows an ExternalAccess-ConsoleAnalyzer, type External access, region United States (Oregon), and a redacted organization zone of trust. The captured status is Creating.

Figure 4 - External Access Analyzer Configured in N. Virginia



The N. Virginia evidence shows Resource analysis - External access selected, region United States (N. Virginia), and Current organization selected as the zone of trust. Analyzer identifiers are redacted.

Figure 5 - Organization Analyzer Reports Active Findings

The resource analysis evidence shows the current organization as the zone of trust and 17 resources with active findings. The analyzer identifier is redacted.

EVIDENCE STATUS

Email notification is documented from the user's completed-lab update and supported by the configured Security Hub alerting pipeline. A separate email screenshot was not supplied for this project.

Security Analysis

Delegated administration reduces management-account usage and centralizes Access Analyzer ownership in the SecurityAudit account.

External access analyzers evaluate supported resource policies and identify access paths outside the selected zone of trust.

Using the organization as the zone of trust is appropriate for this lab because the main concern is public exposure and sharing to external AWS accounts.

Access Analyzer is a detection control, not an enforcement control. It identifies exposure but does not automatically block or remove access.

Findings are not guaranteed to appear in real time. Policy-change analysis can take time, and periodic scans provide additional coverage.

Multi-region deployment is necessary because Access Analyzer analyzers and findings are regional even though IAM is generally treated as a global service.

Lessons Learned

Access Analyzer should be delegated to a security account instead of routinely operated from the management account.

A zone of trust defines which identities or accounts are considered expected for resource sharing.

External analyzers detect public and cross-account exposure across multiple supported resource-policy types.

S3 exposure through ACLs and bucket policies can both generate findings.

Security Hub and email integration turn analyzer output into an operational notification workflow.

Detection should precede broad preventive guardrails in an existing environment.

Production Improvements

- * Deploy organization external-access analyzers in every enabled AWS region.
- * Automate analyzer deployment with CloudFormation StackSets or other organization-level Infrastructure as Code.
- * Create finding-owner and remediation-service-level objectives by resource type and severity.
- * Archive findings only after validating that access is intended and documented.
- * Route findings into ticketing and incident-response workflows in addition to email.
- * Create approved-public and approved-cross-account exception inventories.
- * Monitor analyzer deletion, configuration changes, and delegated-administrator changes with CloudTrail and EventBridge.
- * Use findings to drive Block Public Access, RCP, SCP, and resource-policy remediation.

Skills Demonstrated

Delegated administration	External access analyzers	Organization zone of trust
Multi-region deployment	Active finding review	Security Hub integration
Email alert validation	Data perimeter detection	Remediation planning

Resume Bullet

RESUME	Delegated IAM Access Analyzer administration to a SecurityAudit account, deployed organization-scoped external access analyzers in us-west-2 and us-east-1, identified active resource-sharing findings, and validated centralized Security Hub email notification for data-perimeter exposure monitoring.
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STAR Interview Story

Situation: The organization contained deliberately public S3 resources created through ACL and bucket-policy labs, but lacked a centralized inventory of external resource access.

Task: Deploy an organization-level detection capability without relying on the management account for daily security operations.

Action: I delegated Access Analyzer administration to SecurityAudit, created external access analyzers in Oregon and N. Virginia with the current organization as the zone of trust, reviewed active findings, and confirmed the existing Security Hub email notification path.

Result: The organization gained centralized, multi-region visibility into public and external resource sharing, producing 17 active resource findings in the captured analyzer view and an email alert for analyst follow-up.

Version History

v1.0.0	August 4, 2026	Initial documentation. Delegated IAM Access Analyzer to SecurityAudit, created organization-scoped external access analyzers in Oregon and N. Virginia, documented 17 active resource findings and email alerting, and embedded four supplied evidence screenshots.

References

- * Cloud Security Lab a Week - Sensors Up! Start a Data Perimeter with Access Analyzer.
- * User-provided delegated administrator evidence.
- * User-provided Oregon analyzer evidence.
- * User-provided N. Virginia analyzer configuration evidence.
- * User-provided active findings evidence.
- * Project 43 - S3 ACL Public Bucket Exposure.
- * Project 44 - S3 Misconfigured Bucket Policy Exposure.